Optimal Bayes Classification

Definition

- classification without measurements
- we base just on probability:

$$c_1 \text{ if } P(c_1) > P(c_2)$$
$$c_2 \text{ otherwise}$$

- P_1, P_2 - *a priori* probabilities - our beliefs about the classes **before** seeing the data
- classifier always makes the same decision, depending only on *priori* probabilities
- classification error: $P(e) = \min\{P(c_1), P(c_2)\}$

Probability density function (pdf)

- gęstość prawdopodobieństwa
- we take continuous variable X
- distribution of x values is described by probability density function $p(x)$
- for each interval $\langle x_1, x_2 \rangle$ we calculate probability of X belonging to interval from equation:

$$P(X \in [x_1, x_2]) = \int_{x_1}^{x_2} p(x)\,\mathrm{d}x$$

- distribution funtion:

$$F(x) = P(X < x) = \int_{-\infty}^{x} p(x)\,\mathrm{d}x$$

Class conditional pdf

- $p(x \mid c_i)$
- distribution of feature values for objects of class c_i

Classification using pdf

- we seek probability that object belongs to c_i under condition that feature x was measured:

$$P(c_i \mid x) = \frac{p(x|c_i)P(c_i)}{p(x)}$$

  - $P(c_i \mid x)$ - *posteriori* probability (after observing x)
  - $p(x \mid c_i)$ - how typical is x for class c_i
  - $P(c_i)$ - *prior* probability
  - $p(x)$ - normalising factor for all classes
- object belongs to:

$$c_1 \text{ if } P(c_1 \mid x) > P(c_2 \mid x)$$
$$c_2 \text{ otherwise}$$

- since $p(x)$ is just scaling factor,

$$c_1 \text{ if } p(x|c_1)P(c_1) > p(x|c_2)P(c_2)$$
$$c_2 \text{ otherwise}$$

Loss and Risk

- loss function: $\lambda(\alpha_i \mid c_j)$
  - α_i - decision

  - $i = 1, ..., C$
  - C - amount of decisions
  - c_j - correct class
- expected loss of decision α_i is

$$R(\alpha_i \mid x) = \sum_{j=1}^{C} \lambda(\alpha_i \mid c_j) P(c_j \mid x)$$

Discriminant functions

- for every clas c_i we define discriminant function $g_i(x)$
- dividing feature space into decision regions
- we assign x to c_i if $g_i(x) > g_j(x)$ for all $j \neq i$
- usually for Bayes $g_i(x) = R(\alpha_i \mid x)$ (discriminant function is a function of expected loss of decision)
- boundry between regions is where probability of 2 classes is the same

Multivariate normal distribution

- for multiple dimenshions, variance is replace by **covariance matrix $\mathbf{\Sigma}$**
- diagonal elements are each feature's variance
- off-diagonal elements represent inter-feature covariances

Bayes decision

- for Bayes classifiers with 2D normal distributions and equal priors, key factor is **covariance matrices**
  - if matrices are equal, we have **linear** decision boundry
  - otherwise, we have **non-linear** decision boundry
- in practice, if shapes are equal, just moved in space, decision boundry is **linear**
- when we analize *a priori* probabilities:
  - probabilities are set in order, which area grew the most, and which shrunk
- memory complexity:
  - C - number of classes
  - D - number of features
  - we need **mean** and **variance** for every class and feature plus class priors
  - that gives us $C \times 2D + C = C(2D + 1)$

Estimating pdf with Parzen window

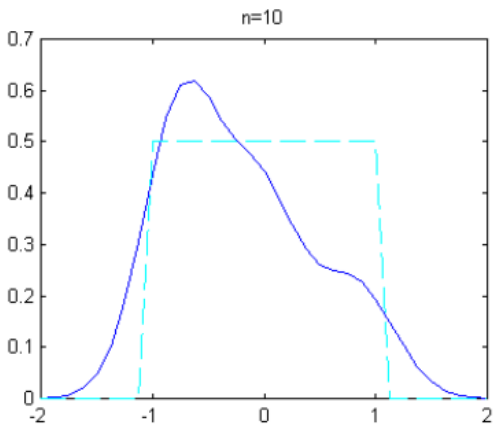
- general idea - "building" unknown dendity from the partial densities of training samples
- **window function** - $\varphi(u)$
  - must be valid density function (non-negative, integral over all domain must be equal to 1)
  - for example simple Gaussian-like window function:

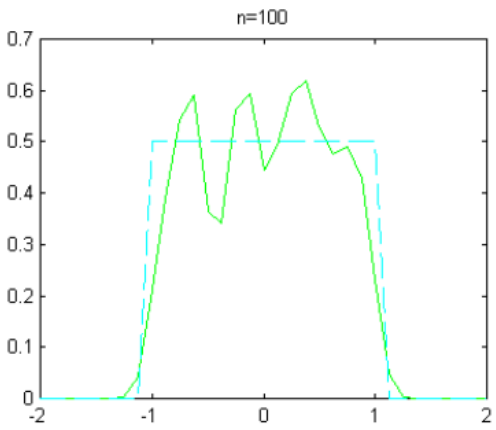
$$\varphi(u) = \frac{1}{\sqrt{2\pi}} e^{-\frac{u^2}{2}}$$

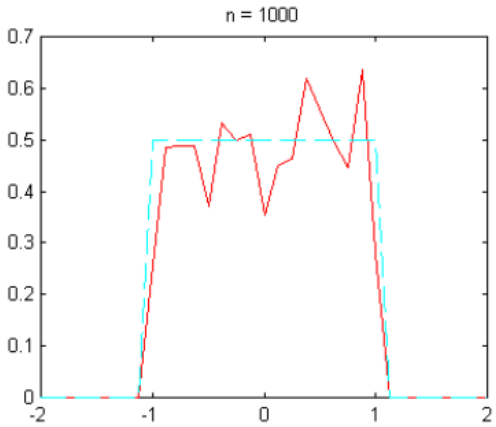
- we often introduce h parameter to scale by number of samples
- pdf estimate is:

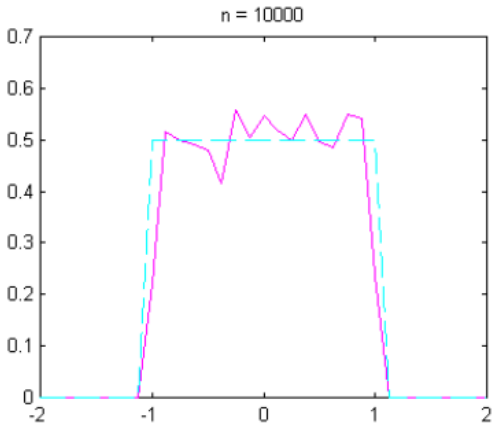
$$p_{n(x)} = \frac{1}{n} \sum_{i=1}^{n} \frac{1}{h_n} \varphi\left(\frac{x - x_i}{h_n}\right)$$

- window with different amount of samples:



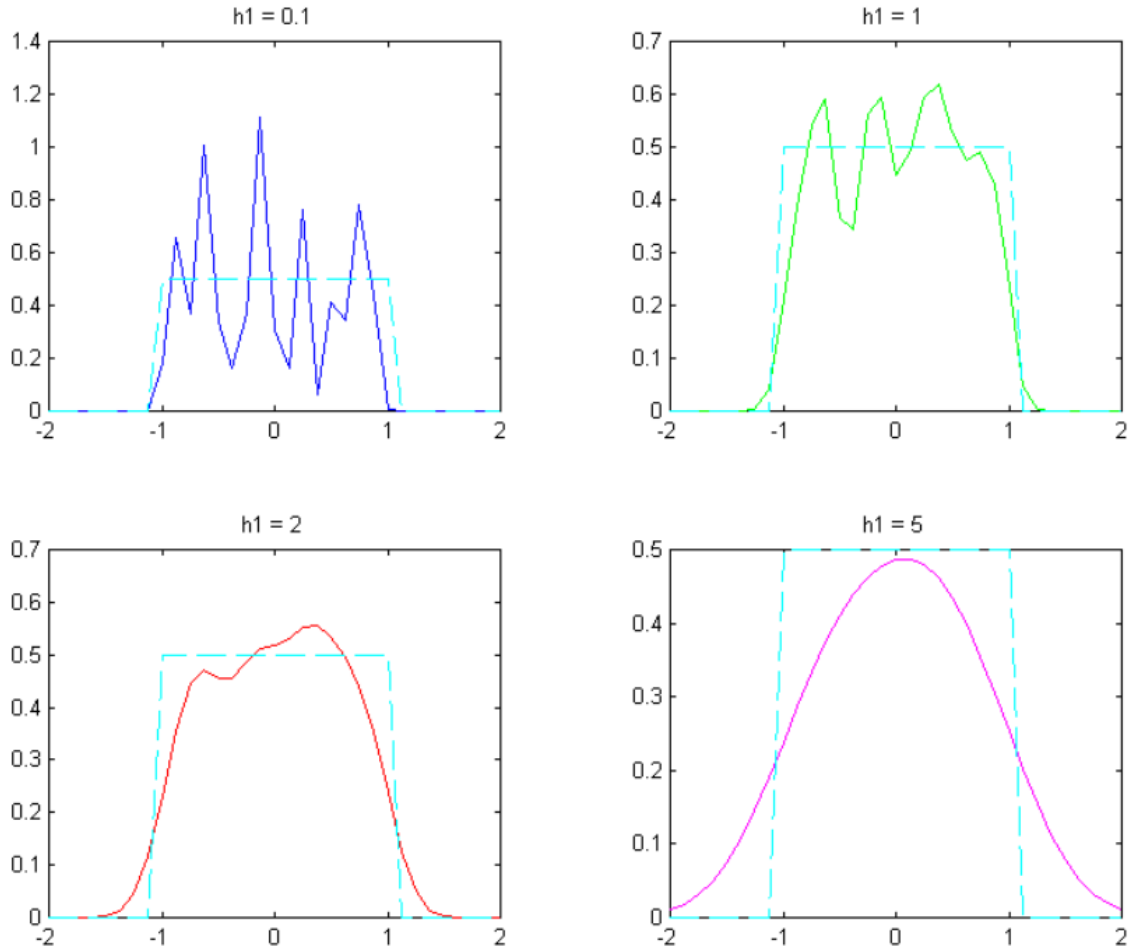






- widths are actually heights (xD):

Different Parzen window *widths*



- memory:
  - for each class classifier must store samples used to create that density ($N \times D$)
  - class labels for samples (N)
  - h parameter
  - final storing: $N(D+1)+1$

Covariance matrix

$$\Sigma = \begin{pmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{pmatrix}$$

- values σ_x^2 i σ_y^2 - how countour is spread in coresponding axises (bigger value - more spread in that diretion)
- value σ_{xy} - in which direction ellipse is tilted:
  - positive - to the right (/)
  - negative - to the left (\)
  - clockwise - positive
  - also, big value - stronger tilt
  - small value - weaker tilt